Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Hydrogen Storage and/or Compression System*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Metal Materials*	*Energy Materials Technologies*
Concept of Proposed Technology	*The proposed R&D project aims to develop and demonstrate a stationary hydrogen storage and/or compression system based on metal-hydride technology, reaching prototype level and near-market readiness. The core concept may lay in the design of a modular system that can be configured either as a hydrogen storage unit, a thermochemical hydrogen compression unit, or a combined storage and compression solution, depending on the specific application requirements. The system will be designed as a flexible product platform that can be adapted to different hydrogen flow rates, pressure levels, and integration scenarios.* *Metal hydrides will be used to reversibly absorb and desorb hydrogen, enabling solid-state storage at low pressure and thermochemical compression without mechanical compressors. Compared to*			

	conventional high-pressure compression systems, the proposed solution offers advantages in terms of intrinsic safety, compactness, low noise, and reduced mechanical complexity. A key feature of the system is the ability to utilize available industrial waste heat for hydride desorption of hydrogen compression, significantly improving overall system efficiency and reducing operating energy costs. *The system will be developed with a strong focus on industrial integration, including interfaces to hydrogen production units such as electrolyzers, hydrogen pipelines, and downstream hydrogen users. By the end of the project, a fully functional prototype system will be manufactured and validated under representative operating conditions, demonstrating readiness for deployment in stationary hydrogen infrastructure applications.*
Rationale for Joint R&D Project	*As hydrogen production and utilization expand across industry and energy systems, there is a growing need for efficient, safe, and cost-effective solutions for hydrogen storage and compression. Current technologies are dominated by mechanical compressors and high-pressure gas storage, which are associated with high electricity consumption, maintenance effort, noise, and safety concerns. These challenges are particularly relevant for decentralized and industrial hydrogen applications where operational efficiency and reliability are critical.* *Metal-hydride-based storage and compression systems offer a promising alternative by enabling solid-state hydrogen handling and thermochemical compression driven by heat rather than electricity. In particular, the ability to utilize low- to medium-temperature waste heat from industrial processes or hydrogen production systems represents a significant opportunity to reduce operating costs and improve system efficiency. However, the development of modular, scalable, and commercially viable metal-hydride systems requires coordinated R&D addressing materials, system engineering, thermal management, and industrial integration.* *A joint R&D project between Fraunhofer IFAM and a Korean SME enables the combination of advanced expertise and IP in metal hydrides and hydrogen systems with industrial engineering capabilities, market knowledge, and commercialization pathways (1-3). Public support is essential to de-risk this technology, enable prototype development, and align system design with real industrial needs, supporting the transition from laboratory concepts to deployable hydrogen infrastructure products.*
Key Words	1. *hydrogen storage* 2. *hydrogen compression* 3. *metal hydride* 4. *waste heat*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6-7)
R&D Objectives	*The primary objective of the project is to develop a modular, metal-hydride-based hydrogen storage and/or compression system as a product-oriented prototype suitable for stationary applications. The project aims to demonstrate reliable hydrogen storage, thermochemical compression, or combined operation with validated performance, safety, and energy efficiency.* *Key objectives include achieving stable and reproducible hydrogen absorption and desorption behavior under application-relevant conditions, demonstrating effective heat-driven compression using industrial waste heat, and validating system scalability through modular design. The project also aims to deliver a system architecture that can be adapted to different use cases, such as buffering hydrogen from electrolyzers, compressing hydrogen for distribution, or enabling decentralized hydrogen storage.* *The project will conclude with a fully integrated prototype system, defined technical specifications, and a commercialization roadmap supporting advancement to TRL 6–7 and preparation for market deployment.*			
R&D Target Specifications	*The metal-hydride-based system will be designed to operate as a stationary hydrogen storage and/or compression unit with low-pressure hydrogen input below 30 bar and hydrogen output pressures up to 30–100 bar, depending on configuration (different pressure values are also possible). The system will target hydrogen storage capacities in the range of several kilograms of active material (up to 1 kg H_2) , scalable through modular replication.* *Thermochemical compression will be driven primarily by heat, with the goal that 70–100 percent of the thermal energy demand for hydrogen desorption can be supplied by available waste heat sources in the temperature range of approximately 60–120 °C. This is expected to significantly reduce electrical energy consumption compared to mechanical compression systems, limiting electricity demand mainly to auxiliary components such as valves, controls, and sensors.* *The system shall demonstrate cycling stability with less than 10 percent performance degradation, and comply with applicable hydrogen safety and industrial standards. The final prototype shall demonstrate safe, reliable operation under representative conditions and achieve TRL 6–7.*			
R&D Plan	*Overall approach and timeline (36 months total)* *The project is structured into two sequential phases, progressing from market analysis and operational definition to manufacturing of pilot-scale module and long-term cycling and reliability testing.*			

Phase I Pre-study and System Definition (6 months)

- *Definition of target stationary applications and operating scenarios, including storage-only, compression-only, and combined system use*
- *Specification of system requirements such as pressure levels, hydrogen throughput, thermal integration, and modularity*
- *Review of applicable Korean hydrogen safety, pressure equipment, and industrial regulations*
- *Selection of suitable metal-hydride materials and preliminary system architecture*
- *Initial assessment of waste heat integration and system energy balance*
- *Preliminary business case and product concept definition*

Milestone Phase I: Approved system concept with defined requirements and go decision for prototype development

Phase II Main R&D, Construction, and Validation (30 months)

- *Detailed design of metal-hydride storage and compression modules including thermal management and safety systems*
- *Manufacturing of pilot-scale modules and system integration*
- *Development of control, monitoring, and safety functions*
- *Experimental validation of hydrogen storage and compression performance, including waste-heat-driven operation*
- *Long-term cycling and reliability testing under representative operating conditions*
- *Definition of modular product variants and preparation of commercialization and scale-up strategy*

Milestone Phase II: Validated prototype metal-hydride-based hydrogen storage and/or compression system achieving TRL 6–7

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IFAM will provide scientific and technological co-leadership in metal-hydride materials and system-level hydrogen technology. The institute will support material selection, laboratory-scale testing, and performance characterization, as well as modeling of thermodynamic and thermal behavior. Fraunhofer IFAM will contribute expertise in waste-heat utilization, system safety concepts, and long-term cycling behavior. The institute will also support validation, TRL assessment, and knowledge transfer.*
Expected R&R of Korean SME Partner	*The Korean SME will lead system engineering, mechanical design, manufacturing, and prototype assembly. Responsibilities include integration of storage and compression modules, development of control systems, and adaptation of the modular design to specific customer needs. The SME will also lead industrial testing, regulatory alignment, and preparation of certification documentation. In addition, the SME will be responsible for commercialization planning, market analysis, and definition of scalable product and service offerings.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project will deliver a validated, modular metal-hydride-based hydrogen storage and/or compression system prototype suitable for stationary hydrogen infrastructure. Technologically, it will demonstrate a safe, low-pressure, and energy-efficient alternative to conventional mechanical compressors and high-pressure storage, with particular benefits when integrated with industrial waste heat sources.* *From an economic and strategic perspective, the project will enable new hydrogen infrastructure products with reduced operating costs and improved safety, supporting wider adoption of hydrogen in industrial and energy systems. The modular design will allow flexible deployment across multiple applications, strengthening the competitiveness of the participating SME in the growing hydrogen market. The project will also contribute to national hydrogen strategies, reduce barriers to decentralized hydrogen deployment, and support long-term industrial decarbonization.*

6. References

(1) EP3045910 B1 patent, Messeinrichtung und Messverfahren zur Bestimmung der Menge eines in einem Speicher aufgenommenen Gases an einem porösen Speichermaterial

(2) EP 3 118 511 B1 patent, Speicherelement für Gase

(3) DE 10 2020 204 214 B4 patent, Behälter zum Speichern von Wasserstoff

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization